New Trade Theory: Scale, Variety and Heterogeneous Firms

1MT495 Current Issues in Applied International Economics

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Where We Are

Lecture 1	Lecture 2	Lecture 3	Today
Balance of payments and investment position: the accounting	International flows: which theory explains which item	Neoclassical trade: trade because countries are *different*	New trade theory: trade because of *scale*, even between similar countries

After today you should be able to:

1. explain why similar countries trade similar goods, and measure it with the Grubel–Lloyd index
2. derive the Krugman (1980) equilibrium and the gains from variety
3. explain the home market effect: why larger markets can attract a disproportionate share of firms
4. use the Melitz (2003) model to explain why only a few firms export and how trade raises productivity
5. connect these models to the gravity equation and to current tariff policy

Outline

1. Introduction to New Trade Theory
2. Krugman (1980): Scale and Variety
3. Melitz (2003): Heterogeneous Firms
4. The Gravity Model
5. Policy and Current Issues

Disclaimer: How to Use This Material

- **Scope:** The objective is a broad overview of the main models and ways of thinking in modern international economics.
- **Focus on intuition:** We will emphasise the economic mechanisms before the formal results. This is not a pure mathematics course.
- **Main lecture versus appendix:** the slides explain the main ideas, examples and predictions; the appendix contains the more demanding derivations.
- **Further reading:** the lecture builds on work by Krugman, Helpman, Melitz and other scholars. References and original sources are listed at the end.

Traditional Trade Theories: Foundations and Limits

- **Ricardo:** countries trade because of differences in **relative labour productivity**. Comparative advantage determines what they export.
- **Heckscher–Ohlin:** countries trade because of differences in **factor endowments**, such as labour and capital abundance.
- **What the simplest traditional models have in common:**
  - Perfect competition and constant returns to scale.
  - Trade patterns driven by differences in technology or factor proportions.
  - No independent reason for *otherwise identical* countries to exchange different brands of the same product.

The puzzle

Why do advanced economies both **export and import cars, machinery and other similar goods**, even when they have similar technologies and factor endowments?

The Need for New Trade Theory (NTT)

- **Intra-industry trade:** Germany exports German cars and imports Japanese cars. Countries exchange *different varieties of similar goods*.
- **Differentiated products:** consumers value a choice of brands, designs and features, not just the cheapest identical product.
- **Economies of scale:** designing a model or setting up production is costly. Selling more units spreads fixed costs over more output.
- **Larger markets and competition:** trade lets successful producers reach more buyers and exposes domestic firms to foreign rivals. This can lower average costs and the prices paid by consumers.

Takeaway: Beyond comparative advantage, trade can yield **larger effective markets, more product variety, and lower costs and prices**. We will study the channels separately.

New Trade Theory vs. Intertemporal Trade

- **Intertemporal trade:** a country borrows and lends internationally, exchanging resources *today* for resources *tomorrow*. The main objects are saving, investment and the current account.
- **New trade theory:** countries exchange *different varieties of similar goods* at the same date, even if their technologies and endowments are similar.
- **They answer different questions:** the intertemporal approach explains why a country spends more or less than its income; NTT explains *what goods are exchanged and why*.

Important distinction

Suppose Germany exports $e30$ billion of cars and imports $e30$ billion of cars. **Net car exports are zero, but gross trade is $e60$ billion.**

Takeaway: Balanced trade does not mean an absence of trade.

What Is New Trade Theory? Definition and Core Mechanisms

- Developed in the late 1970s and 1980s by **Paul Krugman, Elhanan Helpman** and others.
- Adds three mechanisms to the traditional models:
  1. **Increasing returns to scale:** producing more of a variety lowers its *average* cost because fixed costs are spread over more units.
  2. **Monopolistic competition:** firms set prices for differentiated products and compete for customers.
  3. **Love of variety:** households benefit from access to more goods, brands and designs.
- **Broad insight:** opening markets allows specialisation, larger potential sales, stronger competition and more consumer choice, even between otherwise similar countries.

Takeaway: Different models isolate different gains: variety (Krugman), firm selection and expansion (Melitz), and sometimes lower markups.

Intra-Industry Trade: Countries Trade Similar Goods

- **Inter-industry trade:** one country exports machinery and imports agricultural products. Different industries are exchanged.
- **Intra-industry trade:** one country exports cars and imports other cars. The products belong to the *same industry*, but are not necessarily identical.
- **Why would this happen?** German and Japanese firms can specialise in different brands, while consumers in both countries prefer a wider selection.
- **The crucial point:** Germany and Japan need not have different opportunity costs for trade in varieties to be useful.

Compare the mechanisms

Traditional models: **different countries** $\rightarrow$ **different goods**.

New trade theory: **similar countries** $\rightarrow$ **different varieties of similar goods**.

Three Key Concepts in New Trade Theory

Economies of scale

Designing a product and setting up a factory involve fixed costs. When a producer sells more units, the cost per unit falls, even if the marginal cost is unchanged.

Monopolistic competition

Firms sell differentiated products, choose prices and compete for customers. Opening trade can intensify competition and shift sales toward more efficient producers.

Product differentiation and love of variety

Consumers gain from foreign brands and more choice. They may also face a lower *overall price index*, even if a particular firm's price stays unchanged.

Takeaway: A larger market offers **scale**, **selection** and **variety** gains; the models tell us when each channel operates.

Case Study: Germany and Japan Automobiles

Imagine two economies with similar technologies and factor endowments.

- **Without trade:**
  - Germans choose among domestic cars (BMW, Mercedes), Japanese consumers among domestic cars (Toyota, Honda).
  - Smaller markets limit sales and the number of models firms can profitably offer. Fixed costs can keep average costs high.
- **With trade:**
  - Each country specialises in varieties and sells them in a **larger market**; consumers can buy both German and Japanese cars.
  - Successful, productive firms can **expand output**, spread their fixed costs over more sales, and take market share from less efficient firms.
  - Greater choice and reallocation toward efficient suppliers can **lower the overall prices consumers pay**.

Takeaway: The broad intuition is **more variety, larger scale for successful firms, and potentially lower prices**. Krugman and Melitz clarify which effects arise and why.

Measuring Intra-Industry Trade: The Grubel–Lloyd Index

Grubel–Lloyd index for industry i:

$$GL_i = 1 - \frac{|X_i - M_i|}{X_i + M_i} \in [0, 1]$$

- $GL_i = 0$: the country only exports or only imports that industry's products.
- $GL_i = 1$: its exports and imports in the industry are equal.
- The aggregate index weights industries by their share in total trade.

Illustrative example

	X	M	GL
Coffee	0	100	0.00
Cars	30	25	0.91
Machinery	40	10	0.40

Caveat: measured intra-industry trade depends on how industries are defined. It can also capture quality differences and production stages.

Takeaway: The index gives a simple way to measure how much an industry simultaneously exports and imports.

Source: Grubel, H., & Lloyd, P. (1975), Intra-Industry Trade; illustrative numbers.

A Second Puzzle: Why Do So Few Firms Export?

United States

- Of 5.5 million firms in 2000, only **4%** exported.
- The top **10%** of exporters accounted for **96%** of US exports.
- Even in manufacturing only 18% of firms exported (2002), and most sold only a small share of their output abroad.
- Exporters are larger, more productive and pay higher wages than non-exporters within industries.

Why it matters

- Simple models often treat firms as identical.
- But **only some firms can cover the costs of exporting**.
- **Melitz (2003)** extends NTT by allowing firms to have different productivity levels.

Takeaway: Krugman explains why similar countries exchange varieties; Melitz helps explain which firms export them.

Source: Bernard, Jensen, Redding & Schott (2007), Journal of Economic Perspectives; Mayer & Ottaviano (2008), The Happy Few, Bruegel.

Krugman's Model (1980): Why Would Identical Countries Trade?

- Imagine two countries with the **same technology**, factor endowments and preferences. Ricardo and Heckscher–Ohlin provide no motive for trade.
- Consumers nevertheless value **different varieties** of similar goods: cars, clothing, machinery or pharmaceuticals.
- Each firm has a **fixed setup cost**, so it is inefficient for every country to reproduce every variety in small quantities.
- Under **monopolistic competition**, each firm makes one differentiated variety and sells to consumers in both countries.

The key insight

Trade lets countries *specialise in varieties* and consumers *buy a wider range* of varieties. Neither comparative advantage nor factor-endowment differences are required.

Assumptions of the Core Krugman Model

- **Countries:** two identical economies, each with labour endowment L and wage normalised to $w = 1$.
- **Technology:** each firm needs f units of labour to set up, plus c units for every unit produced: $\ell = f + cq$.
- **Consumers:** they buy differentiated varieties and prefer more choice; preferences are CES.
- **Market structure:** monopolistic competition. Firms set prices, but free entry eliminates profits in the long run.
- **Trade costs:** initially zero, so a consumer can buy domestic and foreign varieties at the same factory-gate price.

Takeaway: The model deliberately switches off conventional reasons for trade to isolate economies of scale and love of variety.

Consumer Preferences: Why Does Variety Matter?

Consumers derive utility from consumption of n differentiated varieties:

$$\boxed{U = \left(\int_0^n c_i^{(\sigma-1)/\sigma} \, di \right)^{\sigma/(\sigma-1)}}, \qquad \sigma > 1.$$

- σ is the **elasticity of substitution**: it describes how easily consumers replace one variety with another.
- **High σ:** brands are close substitutes; consumers switch readily when a price rises.
- **Lower σ:** varieties are more distinct; consumers value access to additional choice more strongly.

One immediate implication (equal prices)

If each variety costs p and total spending is Y, then $c_i = Y/(np)$ and

$$U = \frac{Y}{p} \, n^{1/(\sigma-1)}.$$

Even with the *same spending*, access to more varieties raises utility.

CES Demand (1/2): How Do Consumers Allocate Spending?

The problem: choose how many units of each variety to buy without exceeding income Y:

$$\max_{\{c_i\}} U \quad \text{subject to} \quad \int_0^n p_i c_i \, di = Y.$$

Result of utility maximisation (derived in Appendix A.1):

$$\boxed{e_i \equiv p_i c_i = \left(\frac{p_i}{P}\right)^{1-\sigma} Y}, \qquad P = \left(\int_0^n p_i^{1-\sigma} \, di\right)^{1/(1-\sigma)}.$$

- A variety with a **higher relative price** receives less consumer spending because $1 - \sigma < 0$.
- The **CES price index** P summarises the cost of buying the entire basket of varieties.

Takeaway: Consumers substitute towards cheaper brands. The CES price index converts nominal spending into real consumption.

CES Demand (2/2): The Demand Faced by One Firm

Divide expenditure on variety i by its price:

$$c_i = \frac{e_i}{p_i} = p_i^{-\sigma} P^{\sigma-1} Y.$$

When one small firm changes its own price, it takes total spending Y and the aggregate price index P as given:

$$\boxed{q_i = K p_i^{-\sigma}}, \qquad K \equiv P^{\sigma-1} Y.$$

- Its demand is **downward sloping**: a higher price sends customers to competing varieties.
- The perceived demand elasticity is constant:

$$-\frac{d \ln q_i}{d \ln p_i} = \sigma.$$

- A 1% rise in its own price reduces its quantity demanded by approximately σ%.

Takeaway: Product differentiation gives some market power; substitution between varieties limits it.

Firm Technology: What Do Economies of Scale Mean?

A firm must first establish its variety and then manufacture each unit:

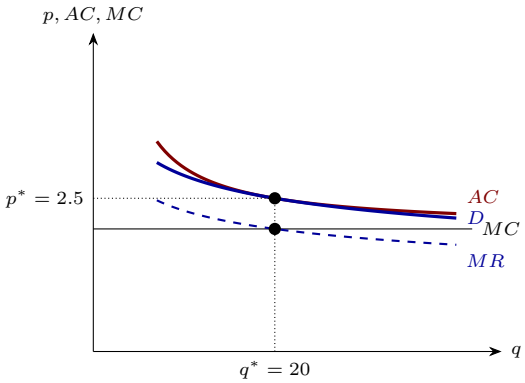
$$\ell = f + cq, \qquad \boxed{AC(q) = \frac{f}{q} + c}, \qquad MC = c \quad (w = 1).$$

- The **fixed cost** f must be paid even if the firm sells only a few units.
- The **marginal cost** c of making one more unit is constant.
- The **average cost** falls with production because the same setup cost is spread over more units.

Illustration: $f = 10$, $c = 2$	$q = 10$	$q = 20$	$q = 40$
Average cost $AC = 2 + 10/q$	3.00	2.50	2.25

Takeaway: Producing each brand at a very small scale is costly. Trade makes it possible to specialise in different brands instead.

Monopolistic Competition: How Prices and Entry Work



1. **Pricing:** a firm produces where $MR = MC$ and reads the price off its demand curve.
2. **Profits:** if price exceeds average cost, more firms enter and split consumer spending across more varieties.
3. **Free entry:** entry ends at $p = AC$: all fixed costs are covered, but economic profit is zero.

Example: $f = 10$, $c = 2$, $\sigma = 5$.

Takeaway: Two distinct forces: firms choose prices optimally, and entry determines how many firms the market supports.

Monopolistic Pricing: Why Is Price Above Marginal Cost?

- Because each variety is differentiated, a firm faces demand $q = Kp^{-\sigma}$ rather than taking the market price as given.
- To sell more it must cut its price; marginal revenue therefore lies below price:

$$MR = p\left(1 - \frac{1}{\sigma}\right).$$

- **Profit maximisation:** set $MR = MC = c$:

$$p\left(1 - \frac{1}{\sigma}\right) = c \quad \Longrightarrow \quad \boxed{p = \frac{\sigma}{\sigma - 1}c}.$$

- **Markup:** $(p - c)/p = 1/\sigma$. The easier it is to substitute across varieties (higher σ), the lower the markup.

Takeaway: Preferences (through σ) determine the markup. Fixed costs determine how much must be sold to break even. Full price FOC: A.3.

Autarky Equilibrium (1/2): Output per Firm

In the long run, price setting and free entry must both hold:

- **Pricing rule:** $p = \dfrac{\sigma}{\sigma - 1} c.$
- **Zero profit:** revenues cover variable and fixed costs, $\pi = (p - c)q - f = 0.$

Combine the two conditions:

$$\left(\frac{\sigma}{\sigma - 1} c - c \right) q = f \quad \Longrightarrow \quad \frac{c}{\sigma - 1} q = f.$$

$$\boxed{q^A = \frac{(\sigma - 1) f}{c}}.$$

Economic interpretation

Each firm must sell enough units to pay the fixed cost out of its markup revenue. Neither market size nor the number of consumers enters this expression.

Autarky Equilibrium (2/2): Number of Varieties

Step 1 – how much labour does one firm need? Substitute its equilibrium output:

$$\ell^* = f + cq^A = f + c\frac{(\sigma - 1)f}{c} = \sigma f.$$

Step 2 – how many such firms fit in the economy? Full employment requires

$$L = n(f + cq^A) = n\sigma f.$$

Therefore the number of firms, and varieties, is

$$\boxed{n^A = \frac{L}{\sigma f}}.$$

- A **larger country** supports more varieties.
- A **higher fixed cost** means fewer varieties because each firm uses more labour.

Takeaway: The number of firms adjusts to market size; equilibrium output *per firm* is fixed by costs and demand elasticity.

Trade Equilibrium: What Changes When Countries Open?

Imagine two **identical** countries, each with L workers. They start with different domestically produced varieties.

1. Without trade, consumers buy only their own country's $n^A = L/(\sigma f)$ varieties.
2. With free trade, the two labour markets jointly support $2L/(\sigma f)$ varieties, and consumers can buy all of them.
3. The firm's markup and zero-profit conditions have *not changed*:

$$\boxed{p^T = p^A, \quad q^T = q^A, \quad n^T_{\text{world}} = 2n^A.}$$

4. Each country produces one set of varieties and imports another: **intra-industry trade** without comparative advantage.

Takeaway: The market doubles the *choice available to consumers*, not the scale or the factory-gate price of each firm in this CES benchmark.

Intuition: Why Trade Raises Welfare in Krugman's Model

- **Consumer gain:** people can now buy foreign brands as well as domestic brands. More variety is valuable even if spending is unchanged.
- **Economies of scale:** each country specialises in a set of varieties rather than duplicating the fixed setup cost of producing every brand at home.
- **What does not happen here:** individual firms do not become larger, and their markups and factory-gate prices do not fall. Those effects arise in other versions of the theory.

With equal prices and symmetric countries, real purchasing power is proportional to $n^{1/(\sigma-1)}$:

$$\boxed{\frac{W^T}{W^A} = \left(\frac{2n^A}{n^A}\right)^{1/(\sigma-1)} = 2^{1/(\sigma-1)}.}$$

Takeaway: In the Krugman (1980) CES benchmark, the gains from opening trade arise from **variety**, not from a decline in firm-level prices.

A Worked Example: Two Identical Countries

Suppose each country has $L = 1{,}000$ workers, $f = 10$, $c = 2$, and $\sigma = 5$.

$$p = \frac{5}{4} \cdot 2 = 2.5, \quad q = \frac{4 \cdot 10}{2} = 20, \quad n^A = \frac{1{,}000}{5 \cdot 10} = 20.$$

	Autarky	**Free trade**
Output per firm	20	20
Factory-gate price	2.50	2.50
Varieties made *in each country*	20	20
Varieties *available to its consumers*	20	40

Consumer real-income gain: $W^T/W^A = 2^{1/4} \approx 1.189$, or about **18.9%**.

Takeaway: Trade lets consumers access twice as many brands without changing output or price per brand.

Home Market Effect: Why the Larger Market Attracts Firms

Now relax one assumption: shipping goods abroad is costly, and one country has a larger domestic market.

- **The basic choice:** a firm can put its factory in either country, but it must pay shipping costs when serving foreign consumers.
- **The attraction:** locating near more buyers saves transport costs on more sales.
- **The counterforce:** the larger market also attracts competitors, who divide local demand among themselves.
- **The result:** under suitable trade costs, the larger country attracts *more than its proportional share* of firms and exports differentiated goods.

A concrete illustration

For example, with 60% of spending in the larger country and trade freeness $\phi = 0.5$, an interior equilibrium assigns **80% of firms** to that country. We focus here on the intuition rather than the full derivation.

Takeaway: Firms locate where customers are, but this depends on shipping costs and competition. It is a location effect, not the basic free-trade variety result.

Melitz (2003): Why Do Only the Best Firms Export?

Return to the Germany–Japan story: which firms actually grow when the market opens?

- In reality, exporting requires **fixed expenses**: distribution networks, regulatory compliance, marketing and finding customers abroad.
- Firms differ in **productivity** φ. A more productive firm has a lower unit cost and can charge a lower price while earning higher operating profits.
- Only sufficiently productive firms find it worthwhile to pay the export fixed cost. These exporters can **increase their sales and scale** by serving additional customers abroad.

Selection and reallocation

Competition from imported varieties reduces demand for some domestic firms. The least productive exit, while productive exporters expand and employ more resources.

Takeaway: Melitz provides the **firm-level expansion and selection mechanism** behind the broader scale-and-cost intuition of the introduction.

Melitz Model: Assumptions

- CES preferences: $U = \left[\int_{\omega\in\Omega} q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{\sigma}{\sigma-1}}$, $\sigma > 1$
- Labour is the only factor; wage $w = 1$; labour force L
- Technology: $l = f + q/\varphi$; productivity φ drawn from $G(\varphi)$, often Pareto
- Entry: sunk cost f_e to draw φ; every period a firm dies with probability δ
- Exporting: iceberg cost $\tau \geq 1$ (ship τ units for one to arrive) and a one-off export entry cost f_{ex}, equivalent to a per-period cost $f_x = \delta f_{ex}$
- Symmetric countries

Prices, revenues, profits

Price: $p(\varphi) = \dfrac{1}{\rho\varphi}, \quad \rho = \dfrac{\sigma-1}{\sigma}$

Revenues:

$$r_d(\varphi) = R\left(P\rho\varphi\right)^{\sigma-1}, \quad r_x(\varphi) = \tau^{1-\sigma}\, r_d(\varphi)$$

Profits:

$$\pi_d = \frac{r_d}{\sigma} - f, \qquad \pi_x = \frac{r_x}{\sigma} - f_x$$

More productive firms charge lower prices and earn higher revenues.

Melitz: Three Hurdles Faced by a Potential Producer

1. Entry lottery

Pay sunk cost f_e to discover productivity φ. At this point, the firm does not yet know whether producing will pay.

2. Domestic survival

Can it cover the operating fixed cost f? If not, it exits after seeing its productivity draw.

3. Export decision

Can it also cover the export cost f_x and ship enough units despite the iceberg cost τ?

Think of a productivity ladder:

$$\varphi < \varphi^* : \text{ exit}; \qquad \varphi^* \le \varphi < \varphi_x^* : \text{ domestic}; \qquad \varphi \ge \varphi_x^* : \text{ export}.$$

Takeaway: Exporters are selected, not randomly chosen: they must clear one more profitability hurdle.

Melitz Model: Who Produces, Who Exports?

Domestic cutoff φ^*: just covers the fixed cost f

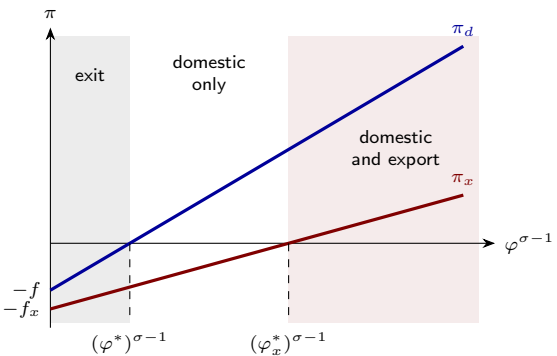
$$\pi_d(\varphi^*) = 0 \;\Leftrightarrow\; r_d(\varphi^*) = \sigma f$$

Firms with $\varphi < \varphi^*$ exit.

Export cutoff φ_x^*: just covers f_x despite the iceberg cost

$$\pi_x(\varphi_x^*) = 0 \;\Rightarrow\; \varphi_x^* = \varphi^* \tau \left(\frac{f_x}{f}\right)^{1/(\sigma-1)}$$

Exporters are more productive than non-exporters if $\tau^{\sigma-1} f_x > f$ – the case supported by the data.



Takeaway: Productivity sorts firms into three groups – exactly the pattern in firm-level data.

Melitz Equilibrium: Two Zero-Profit Conditions with Different Roles

Zero cutoff profit (ZCP)

The least productive *surviving* firm just covers its operating fixed cost:

$$\pi_d(\varphi^*) = 0 \quad \Longleftrightarrow \quad r_d(\varphi^*) = \sigma f.$$

This determines the **survival threshold** for a given market environment.

Free entry (FE)

In autarky, entrants pay f_e before seeing productivity. Expected lifetime domestic profits cover the entry cost:

$$\boxed{\frac{1 - G(\varphi^*)}{\delta}\bar{\pi} = f_e}.$$

Under trade, expected export profits must also enter the free-entry condition.

The Pareto assumption gives an explicit autarky cutoff (Appendix A.7). Under trade, export profits enter free entry; Appendix A.8 derives the export threshold.

Takeaway: ZCP concerns the marginal active firm; FE concerns the expected return *before* the productivity draw.

What Happens When Trade Costs Fall? Follow the Firms

1. **Exporting becomes more attractive.** Existing productive exporters sell more abroad; other productive firms begin exporting.
2. **Competition increases.** Imported varieties win some domestic sales, and weaker firms find it harder to cover fixed costs.
3. **Selection tightens.** The domestic cutoff φ^* rises, forcing low-productivity firms to exit; the export cutoff φ_x^* falls in the benchmark.
4. **Average production becomes more efficient.** Market share and labour shift to high-productivity, often larger exporters, reducing average unit costs across active firms.
5. **Consumers gain.** Better access to foreign varieties and more efficient suppliers lowers the aggregate CES price index in the benchmark.

Important distinction

A particular firm with unchanged productivity still charges its constant-markup price $p(\varphi) = 1/(\rho\varphi)$ (with $w = 1$). Lower *aggregate* prices reflect selection, variety and trade costs—not a fall in every firm's markup.

Takeaway: Trade creates **larger exporting firms and lower average costs** through reallocation, even though some domestic firms contract or exit.

From the Opening Intuition to the Two Models

Recall the Germany–Japan example: trade can bring more choice, larger-scale production and lower prices. The models explain different parts of this story.

	Krugman (1980)	**Melitz (2003)**
Why firms trade	Each produces a distinct variety	Productive firms can pay export costs
Output per firm	Unchanged in the symmetric CES benchmark	Productive exporters expand; weak firms shrink or exit
Firm-level prices	Constant markup and unchanged price	Lower for more productive firms; fixed-φ markup unchanged
Consumer gain	More varieties lower the price index	Variety and selection toward efficient firms lower the price index

Takeaway: The introductory claim is a **broader mechanism**, not a prediction that every firm grows or cuts prices. Melitz adds the reallocation channel missing in the symmetric Krugman benchmark.

The Gravity Equation

Empirical form:

$$X_{ij} = G \frac{Y_i^{\alpha} Y_j^{\beta}}{D_{ij}^{\gamma}}, \qquad \alpha, \beta, \gamma > 0$$

X_{ij} exports from i to j; Y GDP; D_{ij} distance. In logs:

$$\ln X_{ij} = \ln G + \alpha \ln Y_i + \beta \ln Y_j - \gamma \ln D_{ij}$$

- $\alpha, \beta \approx 1$: trade doubles when one partner's GDP doubles
- The **distance elasticity** is $-\gamma \approx -0.9$: doubling distance cuts trade by almost half

Robust facts

- Meta-analysis of 1,467 estimates: mean distance effect 0.91; it has **not** declined since the mid-20th century – no "death of distance"
- **Borders:** in 1988 Canadian provinces traded 22 times more with each other than with US states of similar size and distance (McCallum 1995)
- Accounting for multilateral resistance, borders still cut trade between industrial countries by 20–50% (Anderson and van Wincoop 2003)

Source: Disdier & Head (2008), Review of Economics and Statistics; McCallum (1995), AER; Anderson & van Wincoop (2003), AER.

Gravity: Why Size and Distance Both Matter

Think of a Czech exporter deciding whether to sell in Germany or a distant market.

1. **Market size:** a larger destination has more total spending. That tends to make exports more profitable.
2. **Bilateral costs:** greater distance, border frictions, or tariffs raise delivered prices and reduce demand.
3. **Outside options:** if Germany can buy almost costlessly from many nearby countries, Czech firms face greater competition there. The same bilateral distance can mean something different for an isolated country.

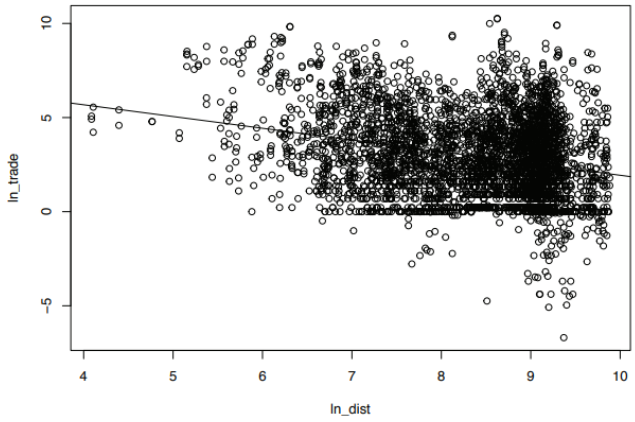
Structural gravity adds the third force

$$X_{ij} = \frac{Y_i E_j}{Y_W} \left(\frac{\tau_{ij}}{\Pi_i P_j} \right)^{1-\sigma}.$$

Π_i and P_j summarize the trade costs of alternative partners (**multilateral resistance**).

Takeaway: What matters is not only the cost of getting from i to j, but that cost relative to trading with everyone else.

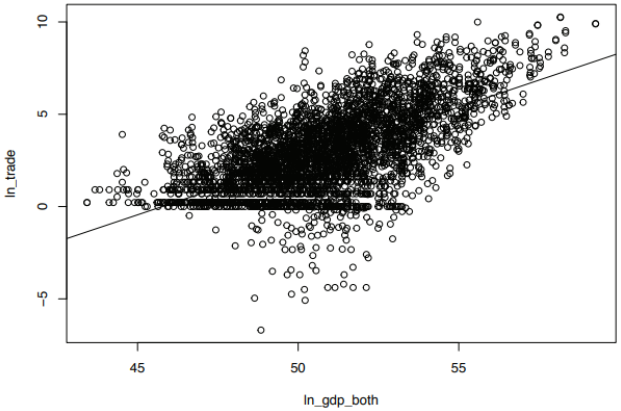
Gravity in the Data: Distance and Trade



Takeaway: Log bilateral trade falls with log distance – the fitted line slopes down, although the scatter is wide.

Source: Shepherd, B., The Gravity Model of International Trade: A User Guide (R version), UNESCAP/ARTNeT,
https://www.unescap.org/sites/default/files/Gravity-model-in-R_1.pdf.

Gravity in the Data: Economic Size and Trade



Takeaway: Log bilateral trade rises with the log of the two partners' combined GDP, with a slope close to one.

Source: Shepherd, B., The Gravity Model of International Trade: A User Guide (R version), UNESCAP/ARTNeT,
https://www.unescap.org/sites/default/files/Gravity-model-in-R_1.pdf.

From New Trade Theory to Gravity

- **Krugman with trade costs:** each variety is made in one place and bought everywhere (love of variety), so trade depends on the size of both markets; iceberg costs τ_{ij}, rising with distance, reduce it
- **Structural gravity** (Anderson and van Wincoop 2003):

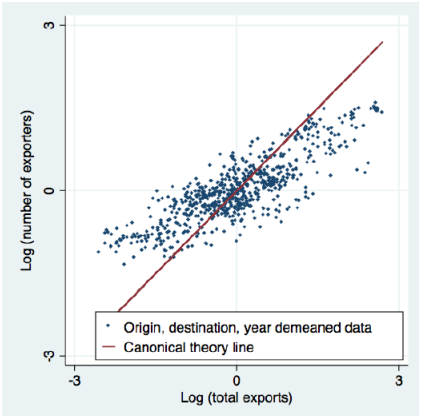
$$X_{ij} = \frac{Y_i\, Y_j}{Y_W} \left(\frac{\tau_{ij}}{\Pi_i\, P_j} \right)^{1-\sigma}$$

  Π_i and P_j are **multilateral resistance** terms: what matters is a pair's trade cost *relative* to their costs with everyone else
- **Melitz with Pareto productivity** (Chaney 2008): higher trade costs also reduce the **number** of exporters, so the trade elasticity is the Pareto parameter $-k$, not $1-\sigma$

Takeaway: Classical models gave gravity no clear foundation; new trade theory does – and turns it into a tool for measuring trade barriers, including tariffs and borders.

How Do Exports Grow? Intensive and Extensive Margins

- **Intensive margin:** average exports per firm
- **Extensive margin:** number of exporting firms (or products)
- **Canonical Melitz–Chaney with Pareto:** a larger destination market raises exports *only* through more exporters – the theory line in the chart has slope one
- **Data** (Fernandes et al. 2023): the slope is clearly below one; the intensive margin accounts for roughly **half** of the variation in exports across markets
- Lower trade costs work through both margins



Takeaway: Exporting more means both more firms and bigger firms – a challenge to the simplest version of the model.

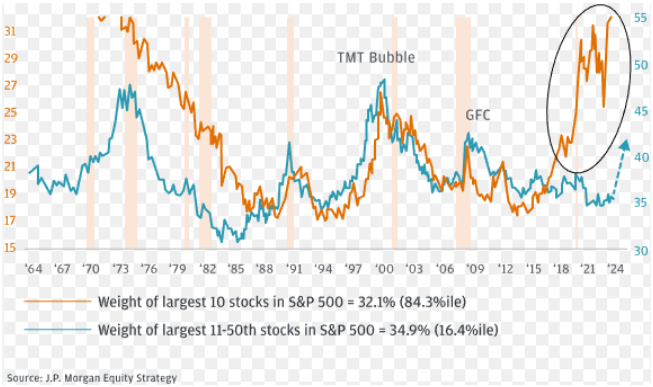
Source: Fernandes, A. M., Klenow, P. J., Meleshchuk, S., Pierola, M. D., & Rodríguez-Clare, A. (2023), "The intensive margin in trade", American Economic Journal: Macroeconomics, 15(3).

Evidence for New Trade Theory

- **Intra-industry trade** is high among advanced economies (cars between Germany and France)
- **Exporters are few, large and productive**, as Melitz predicts
- **Trade liberalisation raises industry productivity** through exit and reallocation (Canada–US FTA, Chile)
- **Home market effects** appear in bilateral trade, especially in differentiated industries with high transport costs
- **Gravity** holds in almost every dataset and has a structural foundation
- **Growth of trade in differentiated goods and services**, where variety and scale matter most

Superstar Firms: An Increasing-Returns Story?

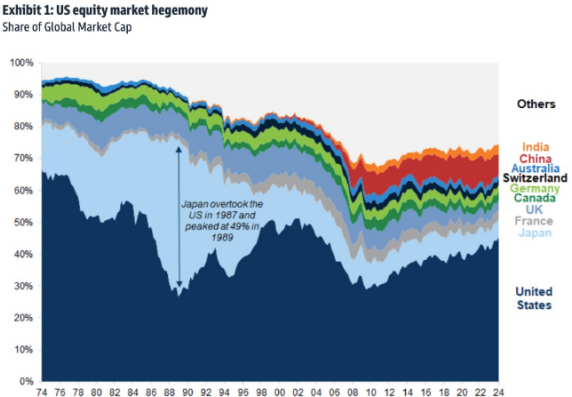
- Increasing returns – fixed R&D costs, network effects, intangible assets – favour large firms; Melitz predicts trade shifts market share to the most productive ones
- The ten largest S&P 500 firms held **37.9%** of the index in June 2026 (peak 40.7% in December 2025), up from 22.8% in 2019
- **Caution:** market capitalisation reflects expected profits and valuations, not market shares; a stock-market chart is suggestive, not a test of the theory



Takeaway: The "superstar firm" evidence (Autor et al. 2020) fits a world of scale economies and winner-takes-most markets.

Source: Chart: J.P. Morgan Equity Strategy; SPDR S&P 500 ETF holdings via ChartRow (June 2026); Autor, Dorn, Katz, Patterson & Van Reenen (2020), Quarterly Journal of Economics.

Concentration Across Countries: The US Share of World Equity



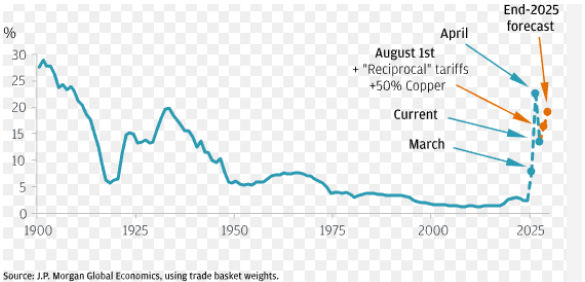
- The US share of global stock market capitalisation fell to about 30% around 1989, when Japan was briefly larger, and has risen since the global financial crisis
- In mid-2026 the United States made up **62.7%** of the MSCI ACWI IMI global equity index (+9.4 pp in ten years)
- **Discussion:** is this a home market effect in technology? Or the result of valuations, buybacks and the dollar?

Source: Chart: Financial Times; MSCI, "A complete geographic breakdown of the MSCI ACWI IMI" (data as of 30 June 2026).

Policy and Welfare Implications

- **Gains from trade** through variety (Krugman), selection and productivity (Melitz), and lower markups (Krugman 1979)
- **Possible costs:** concentration and market power; adjustment costs for exiting firms and their workers; shifts in the location of industry
- **Strategic trade policy** (Brander and Spencer 1985): with large fixed costs and few firms, a subsidy can shift profits to the home firm – the Airbus–Boeing case. But it invites retaliation and requires the government to pick winners
- **Protection in NTT:** a tariff raises τ; the domestic cutoff φ^* falls (weaker selection), the export cutoff φ_x^* rises, and imported variety shrinks – average productivity and welfare fall
- A large country may gain from a tariff by attracting firms (production relocation; Venables 1987, Ossa 2011) – but if partners retaliate, all lose

Current Issue: US Tariffs, 2025–2026



J.P. Morgan estimate and end-2025 forecast, made in 2025

- **2025:** "reciprocal" tariffs under emergency powers (IEEPA) take the effective rate to levels last seen in the 1930s
- **20 Feb 2026:** the Supreme Court strikes down the IEEPA tariffs
- **24 Feb 2026:** a temporary 10% global tariff (Section 122, max. 150 days)
- **24 Jul 2026:** Section 301 tariffs of 10% or 12.5% on 60 economies (99.4% of US imports); sectoral Section 232 tariffs (steel, aluminium, cars) were unaffected by the ruling
- Customs duties were **7.1%** of imports in August 2026, versus 2.3% in January 2025

Takeaway: In NTT terms, tariffs raise τ: fewer exporters (extensive margin), smaller export sales (intensive margin), less imported variety and weaker selection.

Source: J.P. Morgan Global Economics; Penn Wharton Budget Model, "Effective tariff rates and revenues" (14 October 2026); Morgan Lewis (July 2026).

Limitations and Extensions

- **Trade imbalances:** Krugman (1980) and Melitz (2003) are static models with balanced trade; they say nothing about persistent deficits such as the US–China imbalance. For that we need the saving–investment view of lectures 1–2
- **Comparative advantage still matters:** modern models combine Heckscher–Ohlin with scale economies and firm heterogeneity (Helpman and Krugman 1985; Bernard, Redding and Schott 2007)
- **Beyond CES:** variable markups, pro-competitive effects, global value chains, multinational firms
- **Quantitative trade models:** how large are the gains? Arkolakis, Costinot and Rodríguez-Clare (2012) show that in a wide class of models they depend on just two numbers – the share of domestic spending and the trade elasticity

Summary

- **New trade theory complements neoclassical theory:** it explains intra-industry trade between similar countries
- **Driving forces:** increasing returns, product differentiation and monopolistic competition
- **Krugman (1980):** trade multiplies varieties; the real wage rises by $2^{1/(\sigma-1)}$ between two identical countries
- **Home market effect:** with trade costs, a larger market can host more than its proportional share of firms and export differentiated goods
- **Melitz (2003):** only the most productive firms export; liberalisation raises the domestic cutoff and reallocates resources to the best firms
- **Gravity:** trade rises with size and falls with distance (elasticity ≈ -0.9); NTT gives it a structural foundation
- **Policy:** tariffs act as trade costs on both margins of trade – a framework for reading the US tariffs of 2025–26

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- Lecture notes: Isabelle Méjean; Dave Donaldson (`https://dave-donaldson.com/teaching/`); Treb Allen and Costas Arkolakis; Gianluca Benigno (background note on trade and trade imbalances).

Appendix: A Roadmap of the Derivations

One question for each model

- **Krugman (A.1–A.4):** CES demand, pricing, entry and the gains from variety.
- **Melitz (A.5–A.8):** Productivity, the survival cutoff, free entry and the export threshold.

Takeaway: Each proof follows the same sequence: economic question → equilibrium condition → substitution → economic interpretation.

A.1 CES Demand: Why Consumers Buy Every Variety (1/2)

Economic question: How does the price of one variety affect the quantity bought relative to another?

Step 1 – maximise utility subject to expenditure:

$$U = \left(\int_0^n c_i^{\rho} \, di \right)^{1/\rho}, \quad \rho = \frac{\sigma - 1}{\sigma} \in (0, 1), \quad \int_0^n p_i c_i \, di = Y.$$

Step 2 – form the Lagrangian and differentiate for variety i:

$$\mathcal{L} = U - \nu \left(\int_0^n p_i c_i \, di - Y \right), \qquad \frac{\partial \mathcal{L}}{\partial c_i} = U^{1-\rho} c_i^{\rho - 1} - \nu p_i = 0.$$

Step 3 – divide the FOCs for varieties i and j:

$$\left(\frac{c_i}{c_j} \right)^{\rho - 1} = \frac{p_i}{p_j} \quad \Longrightarrow \quad \boxed{\frac{c_i}{c_j} = \left(\frac{p_i}{p_j} \right)^{-\sigma}}, \qquad \frac{1}{\rho - 1} = -\sigma.$$

Takeaway: An expensive variety sells less; σ measures how easily consumers substitute between varieties.

A.1 CES Demand: Use the Budget to Find Absolute Demand (2/2)

Relative demand is not enough: the budget determines *how much* of every variety is purchased.

Step 4 – express all quantities using variety j:

$$c_i = c_j \left(\frac{p_i}{p_j} \right)^{-\sigma} = c_j p_j^{\sigma} p_i^{-\sigma}.$$

Step 5 – substitute into the expenditure constraint:

$$Y = \int_0^n p_i c_i \, di = c_j p_j^{\sigma} \int_0^n p_i^{1-\sigma} \, di.$$

Step 6 – define the CES price index and solve:

$$P \equiv \left(\int_0^n p_i^{1-\sigma} di \right)^{1/(1-\sigma)}, \qquad \boxed{c_j = p_j^{-\sigma} P^{\sigma-1} Y}.$$

Multiplying by p_j gives each variety's spending:

$$\boxed{e_j = p_j c_j = \left(\frac{p_j}{P} \right)^{1-\sigma} Y}.$$

Takeaway: Demand depends on the variety's own price, the cost of alternatives (P), and total spending (Y).

A.2 The CES Price Index and Love of Variety

Economic question: How can trade raise welfare if each firm's price stays unchanged?

Step 1 – substitute CES demand into utility. Since $-\sigma\rho = 1 - \sigma$,

$$U = \left[\int_0^n \left(p_i^{-\sigma} P^{\sigma-1} Y\right)^{\rho} di\right]^{1/\rho} = P^{\sigma-1} Y \left(\int_0^n p_i^{1-\sigma} di\right)^{1/\rho} = \boxed{\frac{Y}{P}}.$$

Step 2 – suppose every firm charges the same price p:

$$P = \left(n p^{1-\sigma}\right)^{1/(1-\sigma)} = \underbrace{n^{-1/(\sigma-1)}}_{\text{variety effect}} p.$$

Step 3 – hold nominal income and individual prices fixed:

$$\boxed{U = n^{1/(\sigma-1)} \frac{Y}{p}}, \qquad \frac{\partial \ln U}{\partial \ln n} = \frac{1}{\sigma - 1} > 0.$$

Takeaway: Additional varieties lower the CES cost of living; the effect is largest when varieties are poor substitutes.

A.3 Firm Pricing: First-Order Condition and Intuition (1/2)

Economic question: How can a firm with a differentiated variety charge above marginal cost? Why does the fixed cost not enter its price rule? **Step 1 – cost and consumer demand (from A.1):**

$$\ell = f + cq, \quad MC = c, \qquad q = Kp^{-\sigma}, \quad K \equiv P^{\sigma-1}Y.$$

Step 2 – write profits as a function of the firm's chosen price:

$$\pi(p) = (p-c)Kp^{-\sigma} - f = Kp^{1-\sigma} - cKp^{-\sigma} - f.$$

Step 3 – differentiate term by term:

$$\frac{d\pi}{dp} = K(1-\sigma)p^{-\sigma} + \sigma cKp^{-\sigma-1} = 0.$$

Step 4 – multiply by $p^{\sigma+1}/K$ and collect terms:

$$(1-\sigma)p + \sigma c = 0 \quad \Longrightarrow \quad \boxed{p = \frac{\sigma}{\sigma-1}c}.$$

Takeaway: Higher prices raise revenue per unit but lose customers. The fixed cost f affects *entry*, not the pricing FOC.

A.3 Free Entry: Why Firm Output Does Not Grow With Trade (2/2)

Economic question: With a larger market, why do we get more *firms* rather than bigger firms?

Step 4 – free entry eliminates profits:

$$0 = \pi = (p-c)q - f = \frac{c}{\sigma-1}q - f \quad \Longrightarrow \quad \boxed{q^* = \frac{(\sigma-1)f}{c}}.$$

Step 5 – substitute the output into the cost and revenue equations:

$$\ell^* = f + cq^* = \sigma f, \qquad pq^* = \sigma f, \qquad AC(q^*) = \frac{f}{q^*} + c = p.$$

Step 6 – verify tangency in the main-slide diagram:

$$AC'(q^*) = -\frac{f}{(q^*)^2} = -\frac{c^2}{(\sigma-1)^2 f}, \qquad \left.\frac{dp}{dq}\right|_{q^*} = -\frac{p}{\sigma q^*} = -\frac{c^2}{(\sigma-1)^2 f}.$$

Takeaway: The markup and fixed cost pin down the scale of each firm. Entry absorbs any increase in market size.

A.4 From Market Size to Welfare: The Complete Chain

Economic question: What exactly changes when two identical markets open to trade? **Step 1 – labour clearing in autarky:**

$$L = n^A(f + cq^*) = n^A \sigma f \quad \Longrightarrow \quad \boxed{n^A = \frac{L}{\sigma f}}.$$

Step 2 – with free trade, world labour doubles but the individual firm's conditions do not:

$$p^T = p^A, \quad q^T = q^A, \quad n^T = \frac{2L}{\sigma f} = 2n^A.$$

Step 3 – lower cost of living comes from the CES index (A.2):

$$P = n^{-1/(\sigma-1)} p \quad \Longrightarrow \quad \boxed{\frac{W^T}{W^A} = 2^{1/(\sigma-1)}}.$$

Step 4 – connect the result to the trade share:

$$\lambda = \frac{1}{2}, \quad \varepsilon = \sigma - 1 \quad \Longrightarrow \quad \boxed{W^T/W^A = \lambda^{-1/\varepsilon}}.$$

Takeaway: First pin down firm size (price setting and entry), then count varieties (resources), and only then compute welfare (consumer prices).

A.5 Melitz: How Productivity Changes a Firm's Outcomes (1/2)

Economic question: If every firm faces the same demand elasticity, why do the most productive firms dominate sales?

Step 1 – cost and price. A firm with productivity φ has marginal labour requirement $1/\varphi$. With $w = 1$ and $\rho = (\sigma - 1)/\sigma$,

$$MC(\varphi) = \frac{1}{\varphi}, \qquad \boxed{p(\varphi) = \frac{1}{\rho\varphi}}.$$

Step 2 – insert the price into CES revenue:

$$r_d(\varphi) = R\left(\frac{p(\varphi)}{P}\right)^{1-\sigma} = \boxed{R(P\rho\varphi)^{\sigma-1}}.$$

Step 3 – variable profit is a fixed fraction of revenue:

$$\pi_d(\varphi) = \underbrace{\frac{r_d(\varphi)}{\sigma}}_{\text{variable profit}} - f, \qquad \frac{r_d(\varphi_1)}{r_d(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma-1}.$$

Takeaway: High productivity lowers price and raises revenue disproportionately when demand is elastic.

A.5 Melitz: How to Aggregate Heterogeneous Firms (2/2)

Economic question: What is the correct "average" productivity for an industry in which high-productivity firms sell more?

Step 4 – condition on firms that operate:

$$h(\varphi \mid \varphi^*) = \frac{g(\varphi)}{1 - G(\varphi^*)}, \qquad \varphi \geq \varphi^*.$$

Step 5 – define a CES-weighted average, not the arithmetic mean:

$$\boxed{\tilde{\varphi}^{\sigma-1} = \int_{\varphi^*}^{\infty} \varphi^{\sigma-1} h(\varphi \mid \varphi^*)\, d\varphi}.$$

Step 6 – in autarky, substitute firm prices into the aggregate price index:

$$\begin{aligned} P^{1-\sigma} &= M \int_{\varphi^*}^{\infty} \big(1/(\rho\varphi)\big)^{1-\sigma} h(\varphi \mid \varphi^*)\, d\varphi \\ &= M\rho^{\sigma-1}\tilde{\varphi}^{\sigma-1} = \boxed{M\, p(\tilde{\varphi})^{1-\sigma}}. \end{aligned}$$

In autarky, the same aggregator gives $R = M r_d(\tilde{\varphi})$ and $\Pi = M\pi_d(\tilde{\varphi})$.

Takeaway: Selection matters because the CES index puts more weight on the high-productivity, high-revenue firms.

A.6 Melitz: Two Equilibrium Conditions, Two Questions

Zero cutoff profit (ZCP): Which firm is just productive enough to operate?

$$\pi_d(\varphi^*) = 0 \iff r_d(\varphi^*) = \sigma f.$$

Since $r_d(\tilde{\varphi})/r_d(\varphi^*) = (\tilde{\varphi}/\varphi^*)^{\sigma-1}$,

$$\boxed{\bar{\pi} = f\left[\left(\frac{\tilde{\varphi}}{\varphi^*}\right)^{\sigma-1} - 1\right].}$$

Free entry (FE), autarky: Is entry worth paying for? A firm survives each period with probability $1-\delta$. Without discounting, expected lifetime *domestic* profits are $\pi_d(\varphi)/\delta$:

$$\boxed{\frac{1-G(\varphi^*)}{\delta}\,\bar{\pi} = f_e.}$$

Interpretation

ZCP determines the marginal domestic survivor. FE compares expected profits *before the draw* with entry costs. With exporting, FE also includes expected export profits.

A.7 Pareto Productivity: Deriving the Domestic Cutoff

Assume $G(\varphi) = 1 - \varphi^{-k}$, $g(\varphi) = k\varphi^{-k-1}$ for $\varphi \geq 1$, with $k > \sigma - 1$ and $\varphi^* \geq 1$.

Step 1 – explicitly integrate the CES productivity moment:

$$\tilde{\varphi}^{\sigma-1} = \frac{1}{1 - G(\varphi^*)} \int_{\varphi^*}^{\infty} \varphi^{\sigma-1} \, k\varphi^{-k-1} \, d\varphi$$

$$= (\varphi^*)^k \frac{k(\varphi^*)^{\sigma-1-k}}{k - \sigma + 1} = \boxed{\frac{k}{k - \sigma + 1}(\varphi^*)^{\sigma-1}}.$$

Step 2 – ZCP implies average profits are constant:

$$\bar{\pi} = f\left[\frac{k}{k - \sigma + 1} - 1\right] = \boxed{f\frac{\sigma - 1}{k - \sigma + 1}}.$$

Step 3 – insert this into FE:

$$(\varphi_A^*)^{-k} \frac{\bar{\pi}}{\delta} = f_e \quad \Longrightarrow \quad \boxed{(\varphi_A^*)^k = \frac{(\sigma - 1)f}{(k - \sigma + 1)\delta f_e}}.$$

Takeaway: The Pareto assumption turns the two equilibrium conditions into a transparent closed-form cutoff.

A.8 Melitz: Why Only Productive Firms Export

Economic question: Why does a firm that survives domestically not necessarily export?

Step 1 – shipping and a fixed export cost reduce foreign profits:

$$r_x(\varphi) = \tau^{1-\sigma} r_d(\varphi), \qquad \pi_x(\varphi) = \frac{r_x(\varphi)}{\sigma} - f_x.$$

Step 2 – export cutoff: set export profits equal to zero:

$$\tau^{1-\sigma} r_d(\varphi_x^*) = \sigma f_x.$$

Step 3 – divide by the domestic cutoff condition $r_d(\varphi^*) = \sigma f$:

$$\tau^{1-\sigma} \left(\frac{\varphi_x^*}{\varphi^*} \right)^{\sigma-1} = \frac{f_x}{f} \implies \boxed{\varphi_x^* = \varphi^* \tau \left(\frac{f_x}{f} \right)^{1/(\sigma-1)}}.$$

Three groups of firms

Exit if $\varphi < \varphi^*$; domestic-only if $\varphi^* \leq \varphi < \varphi_x^*$; export if $\varphi \geq \varphi_x^*$. This ordering requires $\tau^{\sigma-1} f_x > f$.